Andy Widjaja

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Summary

Building production AI systems that improve themselves under human oversight. Tech lead for enterprise AI transformations, most recently an AI SDLC transformation for 8,000–9,000 builders at a Fortune 50 health insurer. At AWS, built and shipped a production agentic system (5,000+ global active users, 90%+ org adoption) and published an open-source MCP server (170,000+ total installs). Background spans sub-millisecond trading gateways (150K contracts/day) to enterprise platforms (800M documents). Building Patina, an open-source cognitive agent framework on Claude Agent SDK with belief graph architecture, graduated autonomy, and MCP integration.

Open Source

Patina: Cognitive Agent Framework (2026) | github.com/Sanctum-Origo-Systems/patina

Cognitive agent framework implementing belief graphs, graduated autonomy, and persistent memory via MCP with three-layer architecture: messaging gateway, Claude Agent SDK runtime, and cognitive MCP server. 35,000+ lines Python, 1,400+ tests.

- Created belief extraction pipeline: LLM-based entity/relationship extraction from unstructured messages into a queryable graph
- Implemented and validated graduated autonomy: agent decision authority expands or shrinks per-domain from measured confidence, exercised against 40+ real decisions
- Implemented MCP server exposing 30+ tools for memory, prioritization, profiling, and introspection to maintain portability across different agentic clients

Autoloop: Governed AI Build Pipeline (2026) | github.com/Sanctum-Origo-Systems/autoloop

Config-driven pipeline that triages, decomposes, and implements GitHub issues autonomously under human oversight. One-command setup, language-agnostic, running in production.

- Built an observer that catches features passing CI but not working in production, auditing the builder's output against actual runtime behavior
- Built governed execution model: protected paths prevent the pipeline from modifying its own infrastructure or safety boundaries
- Processes issues one at a time in dependency order to avoid merge conflicts when implementing multiple issues autonomously
- `autoloop init` sets it up on any repo with existing CI, with no framework to adopt
- Measures its own output: 149 autonomous implementations at 68% first-attempt success and $1.44 per PR, tracked in a published eval report

Experience

AWS · AI Architect, Applied AI (Forward Deployed Engineering) | 2022–Present

Agent Infrastructure (2025–2026):

- Built and shipped Delivery Agent, a multi-step agent orchestration system for software delivery workflows (Python, TypeScript, MCP protocol, LLM reasoning). Scaled from 0 to 5,000+ global active users (90%+ org adoption) across 6 regions within 8 months and led LATAM and APJC technical roadshows (10 countries, 900+ engineers) to drive adoption from builders
- Led the Delivery Agent 3.0 release and mobilized the full team behind it: architecture redesign and code rewrite for modularity across four agent harnesses (Kiro IDE, Kiro CLI, Claude Code, Codex), with 3–7x faster design runs. Built a dependency-graph scheduler that starts each parallel sub-agent stage as soon as its inputs are ready (nearly 2x faster on Kiro Crew), scoped agent tool permissions to trusted commands, and closed the release security review's required findings

- Tech lead for Elevance Health's AI SDLC transformation (multi-million-dollar engagement, Fortune 50 health insurer) for 8,000–9,000 builders globally: designed a platform-agnostic agent orchestration architecture (18 ADRs) to standardize AI-assisted delivery across thousands of repositories, and aligned Elevance engineering, Deloitte, and AWS on scope and roles
- Co-authored ATLAS, an automated evaluation framework measuring agent output quality via precision/recall/F1. Deployed as CI/CD quality gate across 2,500+ concurrent users. Peer-reviewed paper accepted to the AMLC 2026 Gen-AI Autonomous Agents Workshop
- Published document-loader MCP server (awslabs, open source): structured content extraction from PDF/DOCX/XLSX/PPTX/images. 170,000+ total downloads. Shipped a path-containment security fix hardening file access against traversal
- Designed S&P Global agent migration architecture: LangGraph agents and MCP tools to AgentCore Runtime with state migration strategy
- Co-authored AWS Machine Learning Blog post on event-driven ambient agents with human approval (AgentCore)

Enterprise AI Systems (2022–2025):

- Shipped AI-powered document intelligence platform for AIG: 800M documents migrated on ECS Fargate with per-document SHA-256 validation and self-healing retries, 165K daily ingestion, multi-region (NAMER/EMEA/APAC). OpenSearch + Textract pipeline with sensitive payment detection (published on the AWS blog)
- Developed production RAG system for DTCC (public-facing): Anthropic LLM, cross-region inference, 60+ concurrent users, financial regulatory compliance
- Built authorization system for FIS Global using Amazon Verified Permissions: 30x cost reduction, multi-tenant RBAC/ABAC

Legal & General Investment Management America · Full Stack Developer | 2018–2022

- Built serverless event-driven portfolio performance calculation platform for quantitative trading team (Lambda, ECS, Kafka)

Performance Trust Capital Partners · Lead Developer | 2013–2018

- Built proprietary bond analytics engines (CMOs, MBS, munis) and ML models for pricing/asset classification. Decomposed legacy monolith into 20+ microservices (Redis, RabbitMQ, Docker)

R.J. O'Brien & Associates · Trading Systems Developer | 2010–2013

- Built CME Globex trading gateway: 25,000 orders/day, 150,000 contracts, sub-second execution (FIX/FAST protocol). Custom routing gateway for Morgan Stanley

Earlier Career: Newedge/Société Générale, Dell, SEI | 2003–2010

Education, Publications & Speaking

BS Computer Science, University of Wisconsin-Madison
Publications: "High-throughput enterprise-scale document migration on AWS" (AWS Migration & Modernization Blog, 2026); "Building ambient agents with Amazon Bedrock AgentCore" (AWS Machine Learning Blog, 2026)
Speaking: re:Invent 2023–2025, LATAM Agentic AI Tour 2026 (4 countries), APJC Agentic AI Tour 2026 (6 countries)